

## Chapter 7: Differentiation

A Level Pure Maths Notes · Rexponential

### Why this chapter matters

This chapter covers the widest range of new material so far: the chain, product and quotient rules for combining functions; differentiating exponential, logarithmic and trigonometric functions; implicit and parametric differentiation; and using all of this to solve real optimisation problems. Treat this chapter as the one to spend the most time on — everything in Chapter 8 (Integration) is built directly on fluency here, since integration is essentially differentiation in reverse.

The throughline for the whole chapter: a derivative is a **rate of change** — geometrically, the gradient of the tangent to a curve at a point. Every technique that follows (chain rule, product rule, and so on) is just a way of computing that rate of change for more and more complicated functions.

### 7.1 The gradient of a curve, and differentiation from first principles

#### Concept

The gradient of a curve at a point is the gradient of the **tangent line** at that point — unlike a straight line, a curve's gradient changes from point to point, so “the gradient of the curve” only makes sense *at a specific x-value*. Differentiation from first principles builds this idea from scratch: take a chord (a straight line) between the point of interest and a nearby second point, find its gradient, then let the second point slide closer and closer to the first, watching what the chord's gradient approaches. That limiting value is the gradient of the tangent — the derivative.

#### KEY FORMULA: DERIVATIVE FROM FIRST PRINCIPLES

- $$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

#### Procedure

To differentiate a specific function from first principles: substitute  $f(x+h)$  and  $f(x)$  into this formula, expand and simplify the numerator (the  $h$ 's in the denominator should cancel with a factor of  $h$  in the numerator), then let  $h \rightarrow 0$  to see what remains.

#### Worked example

##### EXAMPLE 7.1

Use differentiation from first principles to show that if  $f(x) = x^2$ , then  $f'(x) = 2x$ .

**Step 1**  $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h = \lim_{h \rightarrow 0} [(x+h)^2 - x^2] / h$  (M1, setting up the definition).

**Step 2** Expand the numerator:  $(x+h)^2 - x^2 = x^2 + 2xh + h^2 - x^2 = 2xh + h^2$  (M1).

**Step 3** So  $f'(x) = \lim_{h \rightarrow 0} (2xh + h^2)/h = \lim_{h \rightarrow 0} (2x + h)$  (M1, factoring out  $h$  and cancelling — valid since  $h \neq 0$  in the limit process, even though we then let  $h \rightarrow 0$ ).

**Answer** As  $h \rightarrow 0$ , the term  $h$  vanishes, leaving  **$f'(x) = 2x$**  (A1).

### EXAM TIP

Substituting  $h = 0$  too early, before simplifying —  $[f(x+0)-f(x)]/0$  is  $0/0$ , undefined; the algebra (expanding and cancelling a factor of  $h$ ) has to happen first, and only *then* does letting  $h \rightarrow 0$  make sense.

## 7.2 Differentiating polynomials; tangents and normals

### Concept

In practice, you almost never differentiate from first principles — instead you use the **power rule**, which first-principles reasoning (applied to  $x^n$  in general) can be shown to produce. A **tangent** touches the curve at one point and has the same gradient as the curve there; a **normal** is perpendicular to the tangent at that point, so its gradient is the negative reciprocal of the tangent's gradient (from coordinate geometry, section 3.1).

#### KEY FORMULAS: POLYNOMIALS, TANGENTS AND NORMALS

- Power rule: if  $y = x^n$ , then  $dy/dx = nx^{n-1}$  (works for any real  $n$ , not just positive integers — including negative and fractional powers, once you rewrite roots and reciprocals as powers of  $x$ ).
- Differentiate a polynomial term by term; a constant differentiates to 0.
- **Tangent** at  $(x_0, y_0)$ : find the gradient  $m = f'(x_0)$ , then use  $y - y_0 = m(x - x_0)$ .
- **Normal** at  $(x_0, y_0)$ : gradient is  $-1/m$  (provided  $m \neq 0$ ), then use  $y - y_0 = (-1/m)(x - x_0)$ .

### Worked example

#### EXAMPLE 7.2

The curve  $y = x^3 - 4x^2 + 5$  passes through the point where  $x = 2$ . Find the equations of the tangent and the normal to the curve at this point.

**Step 1** When  $x = 2$ :  $y = 2^3 - 4(2^2) + 5 = 8 - 16 + 5 = -3$ , so the point is  $(2, -3)$  (M1).

**Step 2**  $dy/dx = 3x^2 - 8x$  (M1, applying the power rule term by term).

**Step 3** At  $x = 2$ :  $dy/dx = 3(4) - 8(2) = 12 - 16 = -4$ , so the tangent has gradient  **$m = -4$**  (A1).

**Step 4 Tangent:**  $y - (-3) = -4(x - 2) \Rightarrow y = -4x + 8 - 3 \Rightarrow \mathbf{y = -4x + 5}$  (A1).

**Answer Normal:** gradient  $= -1/(-4) = 1/4$ .  $y - (-3) = \frac{1}{4}(x - 2) \Rightarrow y = \frac{1}{4}x - \frac{1}{2} - 3 \Rightarrow \mathbf{y = \frac{1}{4}x - 3.5}$  (A1).

### EXAM TIP

Using the *function's* gradient formula but forgetting to substitute the specific  $x$ -value —  $dy/dx = 3x^2 - 8x$  is a formula for the gradient at *any*  $x$ ; you must evaluate it at  $x=2$  to get the gradient of the tangent *at that particular point*.

## 7.3 Stationary points, second derivative, increasing/decreasing functions

### Concept

A **stationary point** is where the gradient is momentarily zero — the curve is neither rising nor falling there. This includes local maxima (the curve turns from rising to falling), local minima (falling to rising), and points of inflection with zero gradient (rare, but possible — see 7.4). The **second derivative**,  $f''(x)$ , measures how the gradient *itself* is changing, and gives a quick test for which type of stationary point you have.

### Procedure

1. Find  $dy/dx$  and solve  $dy/dx = 0$  to locate the stationary point(s).
2. Find the corresponding y-value(s) by substituting back into the original function.
3. Classify each stationary point using the second derivative: if  $f''(x) > 0$ , it's a **local minimum**; if  $f''(x) < 0$ , it's a **local maximum**; if  $f''(x) = 0$ , the test is inconclusive and you must check the sign of  $dy/dx$  either side instead.
4. A function is **increasing** on an interval where  $dy/dx > 0$  throughout, and **decreasing** where  $dy/dx < 0$  throughout.

### Worked example

#### EXAMPLE 7.3

Find and classify the stationary points of  $y = x^3 - 3x^2 - 9x + 5$ .

**Step 1**  $dy/dx = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$  (M1, factorising).

**Step 2** Setting  $dy/dx = 0$ :  $x = 3$  or  $x = -1$  (A1).

**Step 3** At  $x = 3$ :  $y = 27 - 27 - 27 + 5 = -22$ , giving the point  $(3, -22)$ . At  $x = -1$ :  $y = -1 - 3 + 9 + 5 = 10$ , giving the point  $(-1, 10)$  (M1, A1 for both y-values).

**Step 4** Second derivative:  $d^2y/dx^2 = 6x - 6$  (M1).

**Answer** At  $x = 3$ :  $6(3) - 6 = 12 > 0 \Rightarrow$  **local minimum at  $(3, -22)$** . At  $x = -1$ :  $6(-1) - 6 = -12 < 0 \Rightarrow$  **local maximum at  $(-1, 10)$**  (A1, both classifications).

#### EXAM TIP

Solving  $dy/dx = 0$  correctly but forgetting to substitute back to find the y-coordinate — a stationary point is a *point* (an  $(x,y)$  pair), and an answer that only gives x-values is incomplete.

## 7.4 Points of inflection, convexity and concavity

### Concept

Convexity describes which way a curve bends: a curve is **convex** (or “concave up”) where it curves upward like the inside of a bowl ( $f''(x) > 0$ ), and **concave** (“concave down”) where it curves like an upside-down bowl ( $f''(x) < 0$ ). A **point of inflection** is where the curve switches from one type of bending to the other — i.e. where  $f''(x)$  changes sign. This is easy to confuse with a stationary point, so it's worth being precise: a point of inflection does NOT require the gradient to be zero there (a curve can be getting steeper-then-less-steep while still climbing throughout, for instance).

## Procedure

1. Find  $f''(x)$  and solve  $f''(x) = 0$  to find candidate  $x$ -values.
2. Confirm it's a genuine point of inflection by checking that  $f''(x)$  actually **changes sign** either side of the candidate (if it doesn't change sign, it's not an inflection point, even though  $f''=0$  there).
3. Find the  $y$ -coordinate by substituting back into  $f(x)$ .
4. (Optional, but often asked) State whether the gradient there is zero (a “stationary point of inflection”) or non-zero (a plain point of inflection).

## Worked example

### EXAMPLE 7.4

Find the point of inflection of  $y = x^3 - 3x^2 + 2$ , and determine whether it is a stationary point of inflection.

**Step 1**  $dy/dx = 3x^2 - 6x$ , so  $d^2y/dx^2 = 6x - 6$  (M1).

**Step 2** Setting  $d^2y/dx^2 = 0$ :  $6x - 6 = 0 \implies x = 1$  (A1).

**Step 3** Check the sign change: for  $x$  slightly less than 1 (e.g.  $x=0.9$ ),  $6(0.9)-6 = -0.6 < 0$ ; for  $x$  slightly more than 1 (e.g.  $x=1.1$ ),  $6(1.1)-6 = 0.6 > 0$ . The sign changes from negative to positive, confirming a genuine point of inflection (M1).

**Step 4**  $y$ -coordinate:  $y = 1^3 - 3(1^2) + 2 = 1 - 3 + 2 = 0$ , so the point of inflection is **(1, 0)** (A1).

**Answer** Gradient there:  $dy/dx$  at  $x=1$  is  $3(1)-6(1) = -3 \neq 0$ , so this is **not** a stationary point of inflection — the curve is still decreasing as it passes through, just switching from concave to convex (A1).

### EXAM TIP

Assuming every point where  $f''(x)=0$  is automatically a point of inflection without checking the sign actually changes — for some functions  $f''$  touches zero without crossing (e.g. at a repeated root of  $f''$ ), and that point is not a genuine inflection.

## 7.5 Differentiating exponentials and logarithms

### Concept

The function  $e^x$  has the remarkable property of being its **own derivative** — this is essentially what defines  $e$  (Euler's number) rather than being a coincidence, and it's why  $e^x$ , not other exponential bases, dominates calculus. Because  $\ln x$  is the inverse of  $e^x$ , its derivative follows from that relationship too, and it turns out beautifully simple:  $1/x$ .

**KEY FORMULAS: EXPONENTIAL AND LOG DERIVATIVES**

- $d/dx(e^x) = e^x$  (the defining property)
- $d/dx(e^{kx}) = k \cdot e^{kx}$ , for constant  $k$  (this is really the chain rule from 7.7, previewed here)
- $d/dx(\ln x) = 1/x$ , for  $x > 0$
- $d/dx(\ln(kx)) = 1/x$  (note: the constant  $k$  cancels out! Since  $\ln(kx) = \ln k + \ln x$ , and  $\ln k$  is just a constant that differentiates to 0.)
- $d/dx(a^x) = a^x \cdot \ln a$ , for a general base  $a > 0$  (less commonly needed, but useful to know it exists)

**Worked example****EXAMPLE 7.5**

Differentiate (a)  $y = e^{3x}$ , (b)  $y = \ln(5x)$ .

**Step 1** (a) Using  $d/dx(e^{kx}) = k \cdot e^{kx}$  with  $k=3$ :  $dy/dx = 3e^{3x}$  (M1, A1).

**Answer** (b) Using  $d/dx(\ln(kx)) = 1/x$  with  $k=5$ :  $dy/dx = 1/x$  (M1, A1). (Notice the 5 doesn't appear in the answer at all — this surprises people the first time, but it follows directly from  $\ln(5x) = \ln 5 + \ln x$ , and  $\ln 5$  is just a constant.)

**EXAM TIP**

Writing  $d/dx(\ln(kx)) = 1/(kx)$  by analogy with a chain-rule pattern applied carelessly — the constant genuinely cancels for  $\ln(kx)$  specifically; this is different from, say,  $d/dx(\ln(x^2)) = 2x/x^2 = 2/x$ , where the “inside function” is not just a constant multiple of  $x$ , so the chain rule result doesn't simplify the same way. Always check whether the inside function is linear-in- $x$ -times-a-constant (simplifies) or something more complex (doesn't).

**7.6 Differentiating trig functions****Concept**

The derivatives of  $\sin$  and  $\cos$  are each other, up to a sign — differentiating  $\sin$  gives  $\cos$ , and differentiating  $\cos$  gives negative  $\sin$ . This elegant, cyclical relationship (differentiate four times and you're back where you started) is a direct consequence of the small-angle approximations from 5.10, and it is **only true when angles are measured in radians** — this is the single most important warning in this section.

**KEY FORMULAS: TRIG DERIVATIVES (ALL ANGLES IN RADIANS)**

- $d/dx(\sin x) = \cos x$
- $d/dx(\cos x) = -\sin x$
- $d/dx(\tan x) = \sec^2 x$  (this follows from the quotient rule applied to  $\sin x / \cos x$  — see 7.9)
- With the chain rule (7.7) built in:  $d/dx(\sin(kx)) = k \cos(kx)$ , and  $d/dx(\cos(kx)) = -k \sin(kx)$

## Worked example

### EXAMPLE 7.6

Differentiate (a)  $y = \sin(3x)$ , (b)  $y = \cos(x^2)$ .

**Step 1** (a) Using  $d/dx(\sin(kx)) = k \cos(kx)$  with  $k=3$ :  $dy/dx = \mathbf{3 \cos(3x)}$  (M1, A1).

**Step 2** (b) This needs the full chain rule since the “inside function” ( $x^2$ ) isn't just  $kx$ : let  $u = x^2$ , so  $y = \cos(u)$ .  $dy/du = -\sin(u)$ ,  $du/dx = 2x$  (M1).

**Answer** By the chain rule,  $dy/dx = dy/du \times du/dx = -\sin(u) \times 2x = \mathbf{-2x \sin(x^2)}$  (A1).

### EXAM TIP

Working in degree mode, or treating an angle like  $3x$  as if it were 3 degrees rather than 3 radians — as soon as you're differentiating trig functions, radians are compulsory, not optional; a question phrased in degrees must be converted to radians before you differentiate.

## 7.7 The chain rule

### Concept

The chain rule differentiates a **composite function** — a function inside another function, like  $\sin(x^2)$  or  $(2x+1)^5$ . The idea: if  $y$  changes with  $u$ , and  $u$  changes with  $x$ , then the overall rate of change of  $y$  with  $x$  is the product of the two individual rates — this is the “multiply the rates together” intuition that makes the chain rule formula memorable.

### KEY FORMULA: THE CHAIN RULE

- If  $y = f(u)$  where  $u = g(x)$ , then  $dy/dx = dy/du \times du/dx$ .

### Procedure

In practice: identify the “inner function”  $u$  (what's being substituted into something else), differentiate the outer function with respect to  $u$ , differentiate  $u$  with respect to  $x$ , then multiply the two results together (and substitute  $u$  back in terms of  $x$  at the end).

## Worked example

### EXAMPLE 7.7

Differentiate  $y = (2x + 1)^5$ .

**Step 1** Let  $u = 2x + 1$ , so  $y = u^5$  (M1, identifying the inner function).

**Step 2**  $dy/du = 5u^4$ ,  $du/dx = 2$  (M1, differentiating each piece).

**Step 3** By the chain rule:  $dy/dx = dy/du \times du/dx = 5u^4 \times 2 = 10u^4$  (M1).

**Answer** Substituting back  $u = 2x+1$ :  $dy/dx = \mathbf{10(2x+1)^4}$  (A1). (Check at  $x=1$ :  $dy/dx = 10(3)^4 = 10 \times 81 = 810$ .)

**EXAM TIP**

Differentiating only the “outer” part and forgetting to multiply by the derivative of the inner function — e.g. writing  $\frac{d}{dx}[(2x+1)^5] = 5(2x+1)^4$  and stopping there, omitting the necessary  $\times 2$  from  $\frac{du}{dx}$ . With practice you can do the chain rule “in your head” (differentiate the outside, multiply by the derivative of the inside), but until it's automatic, write out the u-substitution explicitly to avoid this exact slip.

## 7.8 The product rule

### Concept

The product rule differentiates a function that is one expression **multiplied by** another, where neither part is a plain constant — something the power rule alone can't handle, since  $(fg)' \neq f'g'$  in general (a common wrong guess). Instead, both parts contribute: one term where the first factor is differentiated and the second left alone, plus one term where the second is differentiated and the first left alone.

**KEY FORMULA: THE PRODUCT RULE**

- If  $y = uv$ , where  $u$  and  $v$  are both functions of  $x$ :  $\frac{dy}{dx} = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}$ , i.e.  $(uv)' = uv' + vu'$ .

### Procedure

In practice: identify  $u$  and  $v$ , differentiate each separately, then substitute into the formula. It's worth writing out  $u$ ,  $v$ ,  $\frac{du}{dx}$ ,  $\frac{dv}{dx}$  explicitly as a first step before applying the formula, especially while you're still building fluency.

### Worked example

**EXAMPLE 7.8**

Differentiate  $y = x^2 \sin x$ .

**Step 1** Let  $u = x^2$  and  $v = \sin x$  (M1, identifying the two factors).

**Step 2**  $\frac{du}{dx} = 2x$ ,  $\frac{dv}{dx} = \cos x$  (M1).

**Step 3** By the product rule:  $\frac{dy}{dx} = u(\frac{dv}{dx}) + v(\frac{du}{dx}) = x^2 \cos x + \sin x \cdot 2x$  (M1).

**Answer** So  $\frac{dy}{dx} = x^2 \cos x + 2x \sin x$  (A1, often written this way round, with the higher-degree term first).

**EXAM TIP**

Writing  $\frac{dy}{dx} = (2x)(\cos x)$  only — i.e. differentiating both factors and multiplying the derivatives together, as if the product rule were just “differentiate each part.” Both original terms ( $u$  times  $v'$ , and  $v$  times  $u'$ ) must appear in the final answer, added together.

## 7.9 The quotient rule

### Concept

The quotient rule differentiates one function **divided by** another. It can actually be derived from the product and chain rules (by rewriting  $u/v$  as  $u \cdot v^{-1}$ ), but it's given as its own standalone formula because it comes up often enough to be worth memorising directly — and because the quotient rule's formula has a specific sign and ordering that's easy to get backwards if you try to derive it from scratch under exam pressure.

#### KEY FORMULA: THE QUOTIENT RULE

- If  $y = u/v$ , where  $u$  and  $v$  are both functions of  $x$ :  $dy/dx = [v \cdot du/dx - u \cdot dv/dx] / v^2$ .

### Procedure

A memory aid some students use: “low d-high minus high d-low, over low-squared” ( $v$  is “low,”  $u$  is “high”). Note the order matters here (unlike the product rule) — it's  $v \cdot u'$  minus  $u \cdot v'$ , not the other way round.

### Worked example

#### EXAMPLE 7.9

Differentiate  $y = (2x + 1)/(x - 3)$ .

**Step 1** Let  $u = 2x+1$  and  $v = x-3$  (M1).

**Step 2**  $du/dx = 2$ ,  $dv/dx = 1$  (M1).

**Step 3** By the quotient rule:  $dy/dx = [v(du/dx) - u(dv/dx)] / v^2 = [(x-3)(2) - (2x+1)(1)] / (x-3)^2$  (M1, A1 for the correct substitution).

**Step 4** Expand the numerator:  $2x - 6 - 2x - 1 = -7$  (M1, note the  $x$  terms cancel completely).

**Answer** So  $dy/dx = -7/(x-3)^2$  (A1).

#### EXAM TIP

Swapping the order in the numerator — writing  $u(dv/dx) - v(du/dx)$  instead of  $v(du/dx) - u(dv/dx)$  — which flips the sign of the whole answer. If you're unsure, a quick sanity check: for  $y = x/x^2 = 1/x$  (which you already know differentiates to  $-1/x^2$ ), apply the quotient rule and confirm you get the same, correctly-signed, answer.

## 7.10 Implicit differentiation

### Concept

So far, every function has been given **explicitly**:  $y = (\text{some formula in } x)$ . But some curves — a circle, for instance,  $x^2 + y^2 = 25$  — are defined **implicitly**, mixing  $x$  and  $y$  together with no clean way to isolate  $y$  (or where isolating  $y$  would give two separate branches, like  $y = \pm\sqrt{25-x^2}$ ). Implicit differentiation lets you find  $dy/dx$  directly from an equation like this, without first solving for  $y$ .



## Procedure

Differentiate both sides of the equation with respect to  $x$ , term by term:

- A term in  $x$  alone (like  $x^2$ ) differentiates normally.
- A term in  $y$  alone (like  $y^2$ ) differentiates using the chain rule, since  $y$  is itself a function of  $x$ :  $d/dx(y^2) = 2y \cdot dy/dx$ .
- A term with both  $x$  and  $y$  multiplied together (like  $xy$ ) needs the **product rule**, treating  $y$  as a function of  $x$ :  $d/dx(xy) = x(dy/dx) + y(1) = x \cdot dy/dx + y$ .

Once every term is differentiated, collect all the  $dy/dx$  terms on one side and factor it out, then divide to isolate  $dy/dx$ .

## Worked example

### EXAMPLE 7.10

Find  $dy/dx$  for the circle  $x^2 + y^2 = 25$ , and hence find the gradient at the point  $(3, 4)$ .

**Step 1** Differentiate both sides with respect to  $x$ :  $d/dx(x^2) + d/dx(y^2) = d/dx(25)$  (M1).

**Step 2**  $2x + 2y(dy/dx) = 0$  (M1, using the chain rule on  $y^2$ ).

**Step 3** Rearranging:  $2y(dy/dx) = -2x \implies \mathbf{dy/dx = -x/y}$  (A1).

**Answer** At  $(3,4)$ :  $dy/dx = -3/4$  (A1). *(This makes geometric sense: the radius to  $(3,4)$  has gradient  $4/3$ , and the tangent to a circle is perpendicular to the radius, so the tangent's gradient should be the negative reciprocal,  $-3/4$  — a good way to check your answer independently.)*

### EXAM TIP

Differentiating  $y^2$  as simply  $2y$ , forgetting the essential extra factor of  $dy/dx$  from the chain rule — this is the single most common error in this topic; every time you differentiate a term involving  $y$ , a  $dy/dx$  must appear alongside it.

## 7.11 Parametric differentiation

### Concept

In section 3.4–3.5, a curve was defined parametrically:  $x = f(t)$ ,  $y = g(t)$ , with both coordinates depending on a third variable, the parameter  $t$ . To find the gradient  $dy/dx$  of such a curve, you can't differentiate  $y$  with respect to  $x$  directly (since neither is written as a function of the other) — instead, you use the chain rule in reverse, via  $t$ .

#### KEY FORMULA: PARAMETRIC DIFFERENTIATION

- For  $x = f(t)$ ,  $y = g(t)$ :  $dy/dx = (dy/dt) / (dx/dt)$ .

### Procedure

(This follows from the chain rule:  $dy/dt = dy/dx \times dx/dt$ , so dividing through by  $dx/dt$  gives the formula above.) Find  $dy/dt$  and  $dx/dt$  separately by ordinary differentiation with respect to  $t$ , then divide. The result is usually left in terms of  $t$ , unless a specific point (a specific  $t$ -value) is asked for.

## Worked example

### EXAMPLE 7.11

A curve is defined parametrically by  $x = t^2$ ,  $y = t^3 - 3t$ . Find  $dy/dx$  in terms of  $t$ , and hence find the gradient at the point where  $t = 2$ .

**Step 1**  $dx/dt = 2t$ ,  $dy/dt = 3t^2 - 3$  (M1, differentiating each with respect to  $t$ ).

**Step 2**  $dy/dx = (dy/dt)/(dx/dt) = (3t^2 - 3)/(2t)$  (M1, A1).

**Answer** At  $t = 2$ :  $dy/dx = (3(4) - 3)/(2(2)) = (12 - 3)/4 = 9/4 = \mathbf{2.25}$  (A1).

### EXAM TIP

Trying to eliminate  $t$  and convert to Cartesian form first (as in section 3.4), then differentiating — this works in principle but is usually far slower and more error-prone than differentiating with respect to  $t$  directly and using the formula above; save Cartesian conversion for when the question specifically asks for it.

## 7.12 Connected rates of change

### Concept

Connected rates of change problems involve two or more quantities that are changing with time, linked by some geometric or physical relationship (e.g. the volume and radius of an inflating balloon). The key conceptual tool is the chain rule, used to relate a rate you want (like  $dV/dt$ ) to a rate you're given (like  $dr/dt$ ) via a quantity they have in common (like  $r$ ).

### Procedure

1. Write down the relationship connecting the relevant quantities (e.g.  $V = \frac{4}{3}\pi r^3$ ).
2. Differentiate this relationship with respect to the *linking* variable (often  $r$ ) to get, e.g.,  $dV/dr$ .
3. Use the chain rule to connect the rates you actually want:  $dV/dt = dV/dr \times dr/dt$ .
4. Substitute in known values (both the value of the linking variable, and the given rate) to find the answer.

The chain rule here is doing exactly the same job as in 7.7 — “multiply the rates together” — just applied to real-world changing quantities instead of abstract composite functions.

## Worked example

### EXAMPLE 7.12

A spherical balloon is being inflated so that its radius is increasing at a rate of 2 cm/s. Find the rate at which the volume is increasing at the instant when the radius is 5 cm. (Volume of a sphere:  $V = \frac{4}{3}\pi r^3$ .)

**Step 1** We're given  $dr/dt = 2$  cm/s, and want  $dV/dt$  when  $r = 5$  (M1, identifying what's given and what's wanted).

**Step 2**  $dV/dr = 4\pi r^2$  (M1, differentiating the volume formula with respect to  $r$ ).

**Step 3** By the chain rule:  $dV/dt = dV/dr \times dr/dt = 4\pi r^2 \times 2 = 8\pi r^2$  (M1).

**Answer** At  $r = 5$ :  $dV/dt = 8\pi(5^2) = 8\pi(25) = 200\pi \approx \mathbf{628.3 \text{ cm}^3/\text{s}}$  (3 s.f.) (A1).

### EXAM TIP

Substituting  $r = 5$  into  $dV/dr$  too early (before multiplying by  $dr/dt$ ), then not multiplying by  $dr/dt$  at all, or trying to differentiate  $V$  with respect to  $t$  directly without going through  $r$  — always identify the *chain* of dependence ( $V$  depends on  $r$ ,  $r$  depends on  $t$ ) and build the chain-rule product explicitly.

## 7.13 Optimisation problems

### Concept

Optimisation problems ask you to find the maximum or minimum possible value of some quantity (area, volume, cost, time, ...) subject to a constraint. This is where stationary points (7.3) meet real modelling: the strategy is to express the quantity to be optimised as a function of a **single** variable (using the constraint to eliminate any others), then find and classify its stationary points.

### Procedure

1. Define variables for the quantities involved, and write down the constraint (a fixed relationship between them, often from a fixed total, volume, or perimeter).
2. Write down a formula for the quantity you want to optimise (call it, say,  $S$ ), in terms of more than one variable.
3. Use the constraint to eliminate all but one variable, giving  $S$  as a function of a single variable.
4. Differentiate, set  $dS/dx = 0$ , and solve.
5. Use the second derivative (or a sign check either side) to confirm it's the type of stationary point you want (minimum or maximum), and reject any solution that doesn't make physical sense (e.g. a negative length).

## Worked example

### EXAMPLE 7.13

An open-topped box with a square base is to be made with a volume of  $32 \text{ cm}^3$ . Find the dimensions that minimise the surface area of the box (the base plus the four sides, with no lid), and find that minimum surface area.

**Step 1** Let the base have side length  $x$ , and height  $h$ . Volume constraint:  $x^2h = 32$ , so  $h = 32/x^2$  (M1).

**Step 2** Surface area (base + 4 sides, no top):  $S = x^2 + 4xh = x^2 + 4x(32/x^2) = x^2 + 128/x$  (M1, eliminating  $h$  using the constraint).

**Step 3** Differentiate:  $dS/dx = 2x - 128/x^2$  (M1).

**Step 4** Set  $dS/dx = 0$ :  $2x = 128/x^2 \Rightarrow 2x^3 = 128 \Rightarrow x^3 = 64 \Rightarrow x = 4$  (M1, A1).

**Step 5** Check it's a minimum:  $d^2S/dx^2 = 2 + 256/x^3$ , which at  $x=4$  gives  $2 + 256/64 = 2 + 4 = 6 > 0$ , confirming a **minimum** (M1).

**Answer** Minimum surface area:  $S(4) = 4^2 + 128/4 = 16 + 32 = 48 \text{ cm}^2$ , with base side  $x = 4 \text{ cm}$  and height  $h = 32/16 = 2 \text{ cm}$  (A1).

### EXAM TIP

Optimising the wrong quantity, or forgetting to eliminate a variable using the constraint before differentiating — you cannot differentiate a function of two independent variables using ordinary (single-variable) calculus; the constraint step is what makes the problem tractable at this level, and skipping it is the most common reason these questions go wrong.

## Chapter summary

### Key formulas

#### KEY FORMULAS

- Power rule:  $d/dx(x^n) = nx^{n-1}$
- Exponential and log:  $d/dx(e^{kx}) = ke^{kx}$ ;  $d/dx(\ln(kx)) = 1/x$
- Trig:  $d/dx(\sin kx) = k \cos kx$ ;  $d/dx(\cos kx) = -k \sin kx$ ;  $d/dx(\tan x) = \sec^2 x$
- Chain rule:  $dy/dx = dy/du \times du/dx$
- Product rule:  $(uv)' = uv' + vu'$
- Quotient rule:  $(u/v)' = (vu' - uv') / v^2$
- Parametric:  $dy/dx = (dy/dt)/(dx/dt)$
- Stationary points:  $dy/dx = 0$ ; classify with  $d^2y/dx^2$  ( $>0$  min,  $<0$  max); inflection where  $d^2y/dx^2$  changes sign

### Common exam mistakes across this chapter

- Using trig derivative formulas in degree mode — radians only.

- 
- Forgetting the  $\times du/dx$  (or the extra  $dy/dx$ ) when applying the chain rule, including inside implicit differentiation.
  - Muddling the order of terms/signs in the quotient rule.
  - Finding a stationary point's  $x$ -value but not its  $y$ -value, or not classifying it.
  - Confusing a point of inflection with a stationary point — inflection is about  $f''$  changing sign, not  $f'$  being zero.
  - In optimisation, forgetting to reduce to a single variable using the constraint before differentiating, or forgetting to check the second derivative (or physical sense) to confirm max vs min.

**You should now be able to**

- Differentiate from first principles, and find gradients, tangents and normals to a curve
- Find and classify stationary points, and identify points of inflection
- Differentiate exponentials, logs and trig functions, and apply the chain, product and quotient rules
- Differentiate implicitly and parametrically, and solve connected-rates-of-change and optimisation problems